



Tomáš Plhák

PROGRAMMER · GAME DEVELOPER

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Unity & C# ●●●●● Unreal & C++ ●●●●● Godot & GDScript ●●●●● Flutter & Dart ●●●●●

Python · Bash · JS ●●●●●

Czech — Native · English — Professional

Game developer with a Computer Science background from **Charles University** — Faculty of Mathematics and Physics (Matfyz), game development track and **2+ years** of professional experience at **Flying Rat Studio**. Was part of the team that shipped titles on **PS4, PS5, PS VR2, Xbox One, Xbox Series X|S, and Nintendo Switch**. Fluent in both **Unity/C#** and **Unreal Engine/C++**. Passionate about creative problem-solving, connecting the virtual and real world, and building games that are genuinely fun.

See the full interactive version of this CV at: <https://plhacko.github.io/CV/>

CAREER & EDUCATION

present
2025

Travelling & Independent Development

2 months in Sydney, Australia 🇦🇺 · **2 months in Thailand** 🇹🇭 — working remotely on **two independent app projects**.



Step by Step App

Flutter · Dart · Claude

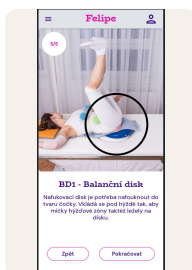
Czech language learning app for foreigners, tied to the successful **Step by Step** textbook series. Also my experimentation with **Claude as a coding assistant**. Initial graphic design by Vojtěch Pinc.

Flutter

Dart

Claude

Early development



Reflex Locomotion App

Flutter · Dart · FlutterFlow · ETA 2026

Medical rehabilitation app for Rehabilitace Karlovy Vary s.r.o., guiding users through the **Vojta Method** (reflex locomotion therapy). **Multi-language support**.

Built with **FlutterFlow** for rapid UI development, with full Flutter/Dart export. Design by Vojtěch Pinc.

Flutter

Dart

FlutterFlow

Medical

Publishing soon

Technology-focused studio specialising in **console porting**, **optimisation**, and **advanced feature development**.



Unannounced Unreal Engine Project

~10 months ▶

Internal studio game built in **Unreal Engine**. Implemented gameplay features in **C++ and Blueprints**, authored **behaviour trees**, integrated audio (SFX hooks), and performed general debugging.

Unreal Engine

C++

Blueprints

Behaviour Trees

My first real **Unreal Engine** project. I worked in both **C++ and Blueprints** — implementing **gameplay systems**, authoring **behaviour trees** for AI, hooking up sound effects at the code level, and performing general debugging across the codebase. The project will be released publicly in the near future.



Beat Saber — PS VR2 Port & Ongoing

~9 months ▶

Part of the Flying Rat team responsible for the **PS VR2 port** of the **world's best-selling VR game**. Later moved to **performance optimisations**, tool creation, and general bug fixing across the codebase.

Unity

C#

PS VR2

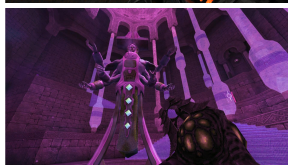
Oculus VR

Internal tooling

Optimisation

Beat Saber is the **world's most popular VR game**. I was part of the Flying Rat team after they delivered the **PS VR2 port** — working in platform-specific VR environment, **performance profiling**, and ensuring the build met **Sony's certification requirements**, performance optimisations, internal tooling, and general bug fixing across the codebase.

[Trailer](#) ↗



Wrath – Age of Ruin — Console Port

~7 months ▶

I was part of the team in porting a boomer shooter from **DarkPlaces** to **Unity**, targeting **PS4, PS5, Xbox One, Xbox Series X|S, and Nintendo Switch**. Responsible for gameplay features, UI implementation, and console-specific bug fixing.

Unity

C#

PS4/PS5

Xbox

Nintendo Switch

Wrath was originally built in **DarkPlaces** — a **Quake engine** derivative. The port involved reconstructing the game in **Unity** from scratch, preserving the feel of the original weapons, movement, and enemy behaviour. My work covered implementing gameplay and UI features and fixing platform-specific bugs across **console targets**, each with their own certification requirements.

[Trailer](#) ↗

Relevant courses:

Programming 1 (Python)

Programming in C#

Programming in C++

Intro to Game Development (Unity)

Introduction to Game Design

Algorithms & Data Structures 1 & 2

Programming 2 (C#)

Advanced Programming in C#

Advanced Programming in C++

Game Mechanics Programming (Godot)

Introduction to Computer Graphics

Practical Game Jam course x4

Projects:



Park Game

Team project · 5 people ▶

Commander-based RTS played in a **real park**. The commander's in-game position is linked to the player's **GPS location**. Designer, project lead, and programmer. Came with a full trailer.

Unity

C#

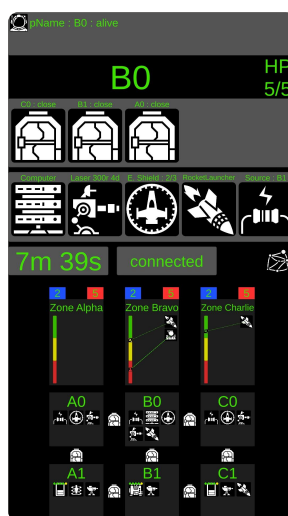
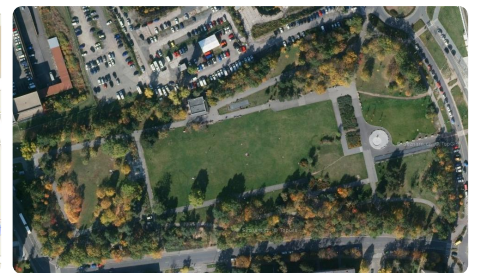
GPS

Multiplayer

Trailer ↗

Gameplay ↗

Players first meet in a park, draw the arena boundaries together, then play a simple **commander-based RTS** where each commander's position is synced to their **real GPS coordinates** — blending the physical world with the game map. I was responsible for the **game design, project leadership**, and parts of the programming. The game shipped with a full gameplay trailer.



Space Alert IRL

Programming in C# · Advanced C# · Annual project ▶

Cooperative **LAN mobile game for Android** inspired by Space Alert. Players move physically between **NFC-tagged rooms** to operate a spaceship. Custom **NFC-Unity integration** documented in a public post.

Unity

C#

Android

NFC

Multiplayer

This project was intended to be my **bachelor's thesis** — I finished the game but started working at Flying Rat Studio before completing the write-up. The game is split into **10-minute missions** where the ship is attacked by enemies and players must use the spaceship's equipment to survive. Moving between rooms requires scanning a **physical NFC tag** placed in that room, connecting the real world with the virtual one. Connecting **NFC and Unity** was unexpectedly complex — I ended up writing a public post documenting exactly how I did it.



PacMan

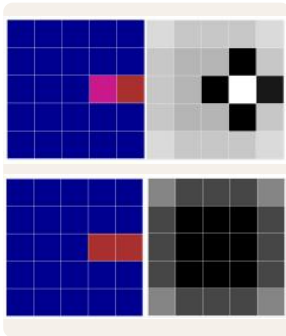
Programming in C++ ►

Custom PacMan with **progressive difficulty** and **authentic ghost behaviours**. **SFML** provides only bare-bones rendering, so all animation systems and text rendering were written **from scratch**. Includes a mouse-driven **map editor** with auto-connecting pipes.

C++

SFML

SFML gives you a window and a sprite batcher — nothing else. I had to build a full **animation state machine**, a **bitmap font renderer** (with variable-width character support), and a **particle system** for the death animations all from scratch. The four ghosts each implement the original Pac-Man AI: **Blinky** directly chases the player, **Pinky** targets four tiles ahead of the player's direction, **Inky** uses a combination of Blinky's position and the player to pick an ambush tile, and **Clyde** alternates between chasing and retreating to his corner. **Difficulty increases over time** — ghosts speed up and power pellet duration shortens with each level. The **map editor** draws the maze with mouse drag and auto-connects pipe segments so they always form a closed, valid playfield.



BattleShips

Programming 1 (Python) · Programming 2 (C#) ►

Two versions: a **Python** console game with adjustable-difficulty AI, then a full **C# WinForms** version with **free-form ship placement** and a **heatmap-based AI** that scores every tile by remaining ship configurations.

Python

C#

WinForms

The **heatmap AI** maintains a **probability grid** that updates after every shot. For each unsunk ship it enumerates every position where that ship could still legally fit — horizontally and vertically — given all known hits and misses. Each candidate position adds weight to the tiles it covers. The AI then fires at the tile with the highest accumulated weight. In **hunt mode** (no active hit streak) this naturally clusters shots around statistically dense areas; once a hit is scored it switches to **target mode** and the distribution collapses tightly around the wound. Ships in the C# version can be placed in any **contiguous non-overlapping shape**, not just straight lines — the AI's enumeration handles **arbitrary polygons**.

High school**Maturita:** Mathematics (state & school), Mathematics+, Computer Science, Czech Language**Volunteer counsellor at YMCA Braník · Lead camp organiser 2022–2023**

Since **age 15** I have volunteered at **summer camps** and at weekly meetings with youth. I regularly **designed and ran** small games — something new **every Friday**. Over the years I also created several larger long-running experiences: two **multi-year board game campaigns** played round by round at weekly meetings, and a large **hex-tile final game** for one of our summer camps. In 2022–2023 I served as **lead organiser** of that same summer camp (YMCA Pecka).

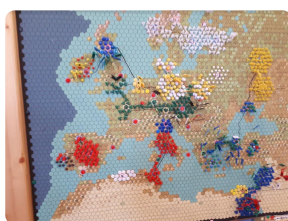
**Dune**

~1 year · Weekly meeting game ▶

Inspired by the **1970s Dune board game**. Players harvest spice on contested hexagons while a **weekly storm** destroys everything in its path. Conflict is central — combat resolved using a system inspired by "**878 Vikings**", where troops may choose to **flee instead of fighting**.

Game Design

Hex Strategy

**Legend**

~2 years · Weekly meeting game ▶

Civilisation V-style strategy played on a **physical wall map**. Players earned points at weekly meetings and spent them colonising territory, building cities, raising armies, and collecting **rare resource sets**. Deliberately designed to create **intrigue** while keeping direct conflict limited. Ran **every Friday** for **nearly two years**.

Game Design

Strategy

**Prince Caspian — Grand Final Camp Game**

2019 ▶

Large-scale hex-tile capture game for a full camp. Teams received **time-limited task cards** gradually released throughout the day — completing them earned captures, defences, or attacks on the shared map. Scoring rewarded **formations and encirclements**, forcing **real-time prioritisation** under time pressure.

Game Design

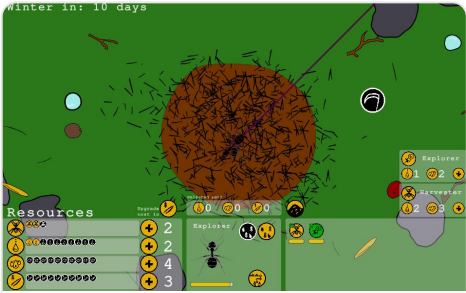
Co-designed with Vojtěch Pinc & Tadeáš Friedrich



Gnome Trouble 3rd / 50

GDS Prague Game Jam 2024 · Theme: Everything changes
2D platformer that switches genre to top-down 3D RPG at the midpoint.

itch.io ↗



Anthill

LUDUM Dare 2023 · Theme: Harvest
Anthill management — deploy scouts and harvesters, upgrade the colony.

itch.io ↗



The Dwarves Delved Too Diligently

Spring Game Jam @ Matfyz 2023 · Theme: Devil is in the details
Grid mining puzzle — regulations pile up after every level.

itch.io ↗



Clumsy Dwarves

GAMEDEV CUNI JAM 2022 · Theme: YOLO
First Unity project. Factory comedy — dwarves die from old age, stupidity, or being crushed by boxes.

itch.io ↗



Bear Breakout: 25D Glass Conundrum

FlyingRat GameJam 2024 · Theme: Fragile
Local co-op for 2 — carry a fragile glass panel across the level.

itch.io ↗



Bold Drive

GDS Prague Game Jam 2025 · Theme: Fortune favours the bold
You're a "Bold" driver (wordplay on Bolt taxis) — drive and chat with passengers.

itch.io ↗

LETTER OF RECOMMENDATION

" I'm happy to recommend Tomáš Plhák, whom I worked with as a Producer/CEO. From the start, Tomáš proved to be a fast learner with a lot of potential. He picked things up quickly, wasn't afraid to ask questions, and consistently showed a genuine interest in improving his skills.

Beyond his work ethic, Tomáš has great social skills and is simply a nice person to be around. He communicates well, works easily with others, and brings a positive, friendly energy to any team. Having him around made collaboration smoother and more enjoyable. Tomáš has a great gift to bring people together both at office and at the conference.

I believe Tomáš would be a great addition to any team, both professionally and personally. I wouldn't hesitate to work with him again and can confidently recommend him for future opportunities. "

Matouš Márai
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